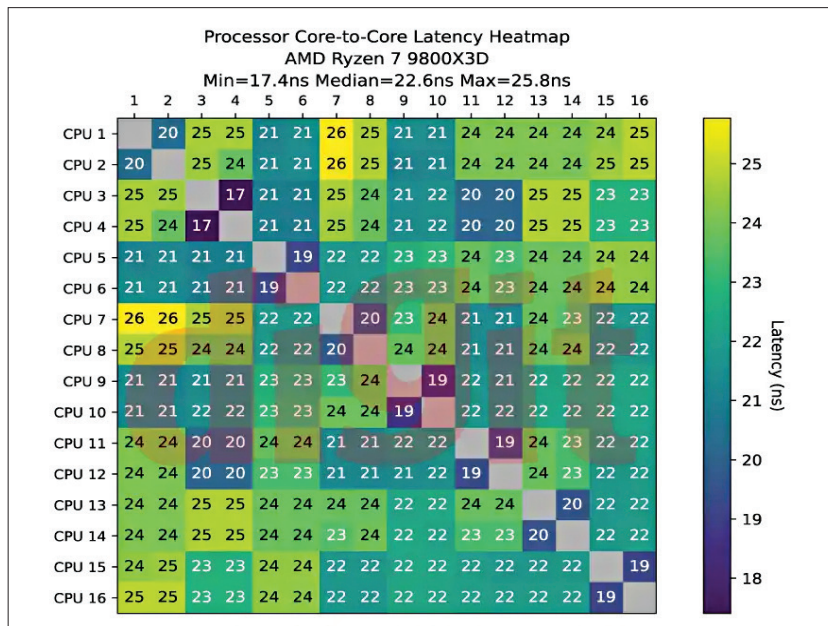
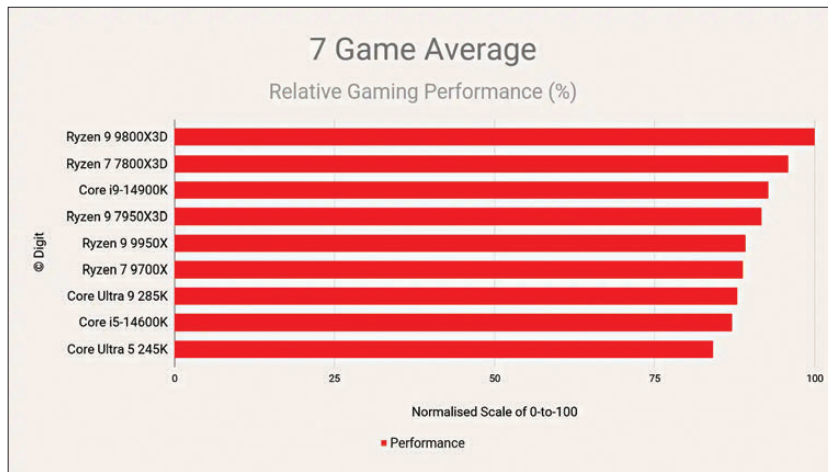




we can see the Ryzen 9 9800X3D shine bright. The stock memory speeds for the AMD CPUs are still a little conservative so in AIDA64, we don't see any stellar performance but knowing that the new 9000 series processors can switch into a 1:1 UCLK:MCLK ratio, we did see the 9800X3D perform quite well. In y-cruncher, we see the Ryzen 9 7950X3D and the Ryzen 9 9950X outperform the 9800X3D. In Web Browser benchmarks, the 9800X3D does manage to do much better and in UL Benchmark's Procyon AI, the Ryzen 9 9800X3D beats the entire stack including the Ultra 9 285K. Coming to the iGPU on the 9800X3D, it's not the best from AMD ever since the 8700G debuted, and even the Ultra 9 285K's new

iGPU completely beats the rest. Finally, coming to gaming, the 9800X3D is a beast. It knocks the socks off all the other processors and dethrones the 7800X3D to become the best gaming processor available right now.

Core to Core Latency – AMD Ryzen 7 9800X3D

Core-to-Core Latency measures the time it takes for data to transfer between different cores within a CPU. It highlights the efficiency of the processor's inter-core communication pathways and cache coherency mechanisms. An interesting insight is that lower core-to-core latency can improve performance in multi-threaded applications where

threads need to synchronize or share data frequently. It can reveal architectural differences between CPUs, such as the impact of chiplet designs versus monolithic die layouts, affecting inter-core communication efficiency.

The AMD Ryzen 7 9800X3D only has 8 Cores so it's easier to keep the median core-to-core latency low and that's definitely beneficial for a lot of computing tasks. Also, it seems like there's only one CCD so there's no added penalty of switching to another CCD. Again, this is also beneficial for quite a lot of workloads.

Power and Thermals

The AMD Ryzen 7 9800X3D peaked at 150 Watts over a 10-minute throttling run. And the average wattage was a little lesser than that. Compared to the 9700X, this is a bit higher but at no time did we ever see any throttling. Even when we swapped out the cooler for a single-fan air cooler, the CPU ran just fine. Average temperatures were at 93 degrees Celsius and peak was at 95 degrees.

VERDICT

AMD has added another successful X3D processor in its kitty with the addition of the AMD Ryzen 7 9800X3D. It's a great performer for everyday productivity, content creation and when it comes to gaming, it knocks it out of the park yet again. The only place where the 9800X3D takes a hit is in the value for money aspect because the price is a little too much to justify the added gains with gaming performance. We'd have preferred it to be a tad above the Ryzen 7 9700X and not this far apart. In India, the USD 120 difference works out to more than INR 12,000 which can be quite discouraging for gamers.

–Mithun Mohandas

SPECIFICATIONS

Cores: 8 | Threads: 16 | Base clock: 4.7 GHz | Boost clock: 5.2 GHz | Process node: TSMC N4 | L3 Cache: 96 MB | L2 Cache: 1 MB per core | TDP: 120 W | Socket: AM5 | Architecture: Granite Ridge

CONTACT

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